

6	(a) (i) progressive wave transfers energy, stationary wave no transfer of energy/ keeps energy within wave	B1	[1]
	(ii) (progressive) wave/wave from loudspeaker reflects at end of tube reflected wave overlaps (another) progressive wave same frequency and speed hence stationary wave formed	B1 B1 B1	[3]
	(iii) (side to side) along length of tube/along axis of tube	B1	[1]
	(b) all three nodes clearly marked with N/clearly labelled at cross-over points	B1	[1]
	(c) phase difference = 0	A1	[1]
	(d) (i) $v = f\lambda$ $\lambda = 330/440 = 0.75\text{ m}$	C1 A1	[2]
	(ii) $L = 5/4 \lambda$ $= 5/4 \times 0.75 = 0.94\text{ m}$	C1 A1	[2]